OpenMP

- Open MultiProcessing, https://www.openmp.org/
- API for shared-memory parallel programming
- usage
  - compiler directives `#pragma omp ...`
  - functions
  - environment variables
- compiling

```
#include "omp.h"

gcc … -fopenmp ...

srun --cpus-per-task=N
```

- Main OpenMP elements
  - library functions
    - `omp_get_thread_num`, `omp_get_num_threads`
  - environmental variables
  - distribution of work
    - directives `parallel`, `parallel sections`, `parallel for`
    - `scheduling` clause
  - synchronization
    - directives `critical`, `atomic`, `barrier`, `single`, `master`
  - scope of variables
    - clauses `shared`, `private`, `reduction`

Some Library Functions

- number of threads
  - `omp_set_num_threads`: sets number of threads to be used in parallel section
  - `omp_get_num_threads`: get currently set number of threads
  - `omp_get_max_threads`: get max. number of threads (on hardware level)
  - `omp_get_thread_num`: get thread index 0 … omp_get_num_threads
- number of processors/cores
  - `omp_num_procs`: number of processors available
- timing
  - `omp_get_wtime`: wall-clock time elapsed from some point in the past
  - `omp_get_wtick`: resolution of wall-clock time measurement

Environmental Variables

- user can influence the behavior of executable code at runtime
- set number of threads

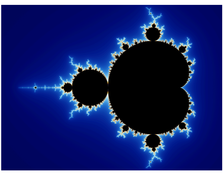
```
export OMP_NUM_THREADS=4
unset OMP_NUM_THREADS
```

Distribution of Work

- directive `parallel`
  - source hello1.c
  - check the printouts
- clauses `sections`, `section`, and `nowait`
  - source hello2.c
  - barrier is automatically set after each sections directive
  - to disable it add `nowait` clause
- `parallel for`
  - most frequently used OpenMP functionality
  - distributes iterations among threads
  - for statement must be given in canonical shape

```
                                ++i
                                i++
                                --i
                        i <     i--
  for (int i = iStart; i <= iEnd; idx += iStep      )
                        i >=      idx -= iStep
                        i >       idx = idx + iStep
                                  idx = idx - iStep
```

- `parallel for collapse (num)`
  - parallelization of `num` nested loops
- `parallel for schedule(type, number)`
  - useful when workload significantly differs from iteration to iteration
  - `type`
    - static: iterations are mapped to threads at compile time, default
    - dynamic: when thread finishes its work, it gets new number of iterations to compute
    - guided: each threads gets larger portion of iteration at the beginning, then the number is reduced
    - runtime: set though environment variables
  - `number`: the smallest chunk of iterations assigned to a thread
- example: The Mandelbrot set
  - an image of The Mandelbrot set from Wikipedia



  - the Mandelbrot set (black) is the set of all points c in the complex plane where the value converges with iterations
    - algorithm

$$z_0 = 0$$
$$z_{k+1} = z_k^2 + c$$
$$\mathrm{count}(c) = \min_{0 \le k < K} (|z_k| \ge 2)$$

    - Divergence for large z
    - Compute the function up to some maximum value K
  - serial control flow in elemental functions
  - load imbalance
  - implementation
    - as opposed to vector operations like saxpy, it cannot be efficiently computed on SIMD systems
    - best with SPMD or tiled SIMD
  - mb0.c: manual parallelization
  - mb1.c: parallelization of outermost loop using the parallel for
  - mb2.c: parallelization of both loops
  - mb2.c: dynamic scheduling improves performance

Synchronization

- dependencies between tasks require synchronization
- overhead
  - launching and synchronizing tasks
  - over-decomposition increases overhead

Race conditions

- concurrent tasks perform read and write operations at the same memory location
- are tricky, should be avoided
- example

```
Task A      Task B
X = 1;      Y = 1;
a = Y;      b = X;
```

  - initial values: `X = 0, Y = 0`
  - possible outcomes:
    - `a = 1, b = 0`
    - `a = 0, b = 1`
    - `a = 1, b = 1`
- critical section: only one thread enters critical section at once
- directives `critical` and `atomic`
  - prevent race condition
  - hardware implementation
    - modern processors do not lock memory bus, but works on cache line
    - works together with the MESI cache coherence mechanism
    - instructions with LOCK prefix
      - synchronize the cache-line with the main memory
      - acquire exclusive access to the cache line and mark it locked
      - perform read-modify-write on operands in the cache line
      - mark the cache line modified and unlocks it
    - while the cache line is locked, the cache coherence requests of other CPUs are ignored
  - performance of directives
    - `critical` uses locks and performs operation in 3 steps:
      - lock (atomically sets a variable),
      - arbitrary number of instructions,
      - unlock
    - `atomic` combines all together – instruction takes care of locking and modifying a variable atomically
      - applicable only to scalar variable assignment with operators ++, –, +=, -=, *=, /=, &=, |=, <<=, >>=
      - faster than critical
- example:
  - computing π following Lebnitz formula

$$\pi = \sum_{i=0}^{N-1} (-1)^i \frac{4}{2i+1}$$

  - pil0.c: does not compile as for loop is not in canonical form
  - pil1.c: loop dependence, race condition
  - pil2.c: correct result, poor performance